

DATA ANALYSIS PROJECT

(European Bank Churn Pattern Analysis)



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Customer segmentation & Churn Pattern Analytics in European Banking

Meaning Of The Project :

The main objective of this project is to analyze customer churn in a European banking dataset. Customer churn refers to customers who have stopped using the bank's services or have left the bank. This project aims to identify the overall churn rate and analyze churn across different customer segments. Additionally, the study examines churn rates based on geographical regions and Age-wish Churn rates also. The insights gained can help the bank develop effective customer retention strategies.

Background and Context :

Customer churn is one of the largest hidden costs in the retail banking industry. High churn rates can have several negative impacts on a bank, including :

- Reduced Life time value.
 - Increased costs for acquiring new customers.
 - Reduced profitability and business growth.
 - Revenue instability and uncertainty.
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Importance of the study : IMPORTANT

Although banks monitor overall churn rates, they often lack detailed customer-level insights that can help them understand the reasons behind customer attrition. This project aims to answer the following key questions:

- Which customer groups are most likely to churn ?
- How do churn rates differ across geographical regions ?
- How does customer age influence churn behavior ?

Problem of the Statement :



Here the Main problem is Customer churn. Firstly analyze the Churn rates after find the customers segmentation wish churn rates.

- Identify High-risk customers segmentation.
 - Understand the Churn rates differs across from the Geography, age groups churns.
 - Quantifying the financial profile the churn customers.
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Project Objectives :



- Measure the Overall Churn rate.
 - Identify churn distribution across from the customer segmentation.
 - Compare churn Behavior across from European regions.
 - To provide recommendations for reducing customer churn and improving customer retention.
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Secondary Objectives :

- Understand churn distribution among the high – value customers.
 - Evaluate engagement and tenure patterns.
 - Support strategic planning and marketing decisions.
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Dataset Description :

Column	Description
CustomerId	Unique customer identifier
Surname	Customer surname
CreditScore	Customer creditworthiness
Geography	France, Spain, Germany
Gender	Male / Female
Age	Customer age
Tenure	Years with the bank
Balance	Account balance
NumOfProducts	Number of bank products
HasCrCard	Credit card ownership
IsActiveMember	Activity indicator
EstimatedSalary	Estimated annual salary
Exited	Churn indicator (target)

Analytical methodology :



Data Cleaning & Preparation Process :

- ✓ Load the CSV dataset into Python using the Pandas library.
- ✓ Examine the dataset and identify unnecessary columns that are not relevant to the analysis. Ex.(CustomerId and Surname)
- ✓ Remove the unnecessary columns from the dataset.
- ✓ Check for missing (null) values and handle them appropriately.
- ✓ Create customer age groups using lambda functions to categorize customers into different age segments.
- ✓ Develop a Financial Profile for each customer by combining key financial attributes such as account balance, number of products, estimated salary, and credit score.
- ✓ Use percentile ranking (rank(pct=True)) to assign scores to each financial attribute and calculate an overall Financial Score.
- ✓ Classify customers into **High Value** and **Low Value** categories based on the Financial Score using a lambda function.
- ✓ Verify the transformed dataset and ensure it is ready for exploratory data analysis and churn analysis.

Customer Segmentation Design :



Segmentation dimensions include:

Geographic Segmentation:

France, Spain, Germany

Age Segmentation:

<30, 30–45, 46–60, 60+

Credit Score Bands:

Low, Medium, High

Tenure Groups:

New, Mid-term, Long-term customers

Balance Segments:

Zero-balance, Low-balance, High-balance

First of all we can see the Overall churn rate and Total number of customers in the dataset.

- **Total Customers** - 10K (100%)
- **Total Churn** - 2037 (20.37%)

Here Analyzing the Customers with the use of segmentation also.

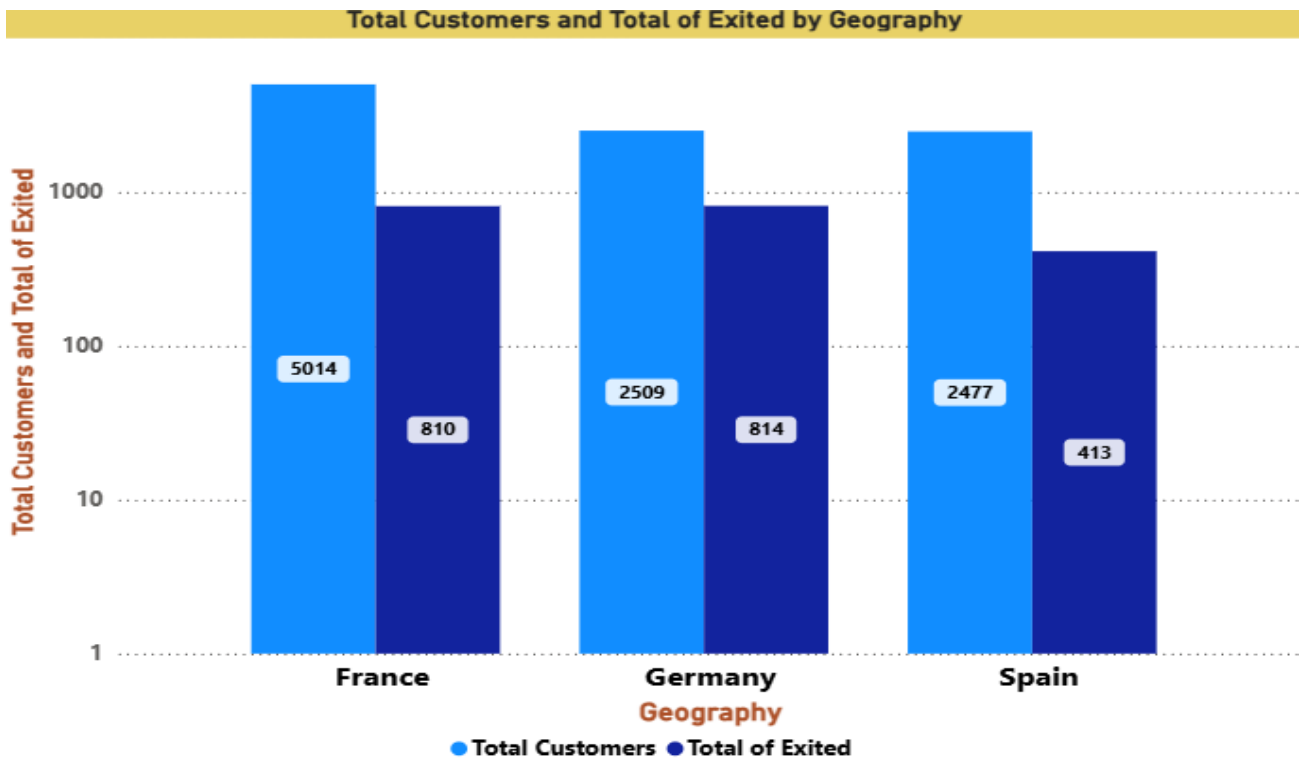
Customer Segmentation :

Geographic Segmentation :

Here the European Bank dataset having the 3 type of Regions. This is all given below,

- France
- Spain
- Germany

Bar Diagram also Given Below,

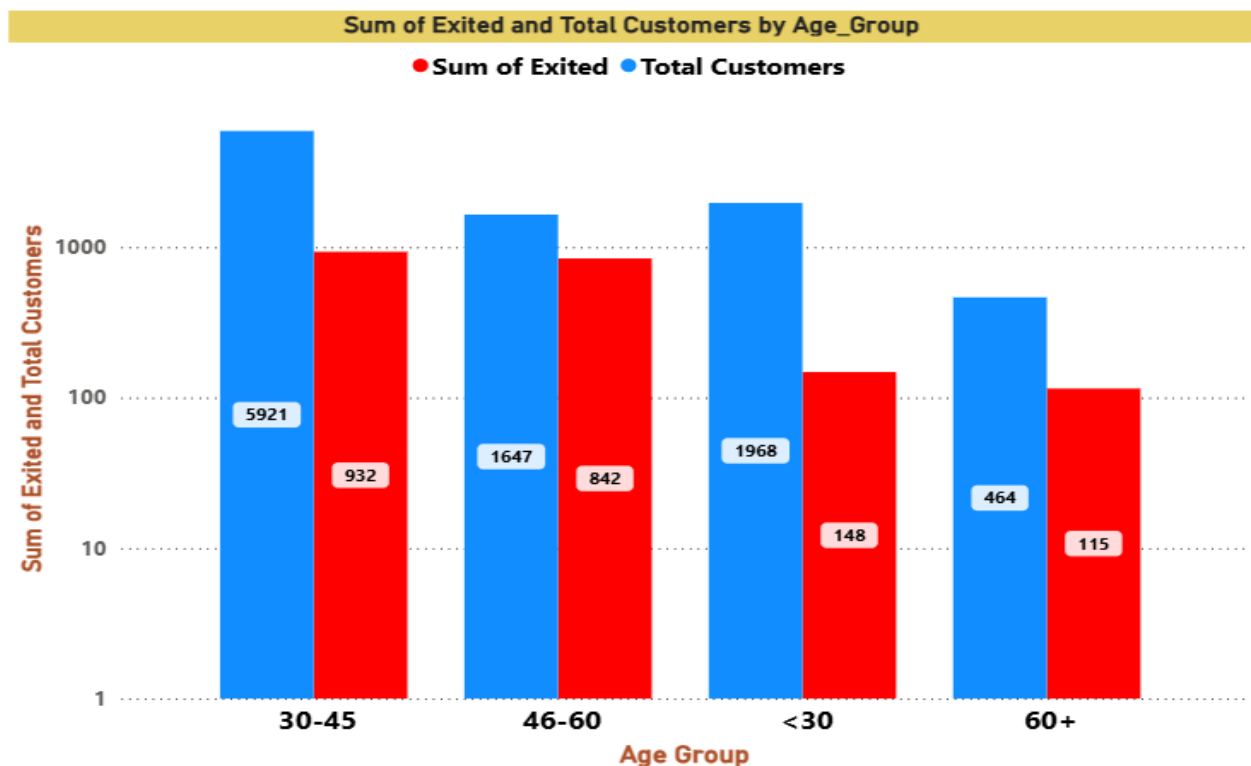


Here in the Diagram shows,

- France's churn rate is **16.16%**, contributing **8.1%** to the overall churn rate of **20.37%**
- Germany's churn rate is **32.44%**, contributing **8.14%** to the overall churn rate of **20.37%**
- Spain's churn rate is **16.67%**, contributing **4.13%** to the overall churn rate of **20.37%**

Age Segmentation :

Age Segment	Total Customers	Exited	Retained	Churn Rate
Below 30	1968	148	1820	7.52%
30–45	5921	932	4989	15.74%
46–60	1647	842	805	51.12%
60+	464	115	349	24.78%



- **30–45 age group has the highest number of churned customers (932)** because it contains the largest number of customers.
- **46–60 age group has the highest churn rate (51.12%)**, meaning more than half of the customers in this age group left the bank.
- **Below 30 age group has the lowest churn rate (7.52%)**, indicating younger customers are more likely to stay.
- The bank should focus retention efforts on customers aged **46–60**, as they represent the highest churn risk.

Geography-Age Interaction Analysis :

Now here we are see the difference between the Age-group churn contribution with separating Geography wise. Below the Bar Diagram explaining the Geography wise Age-segmentation contribution to churn.

○ France Customer Churn Analysis by Age Group :

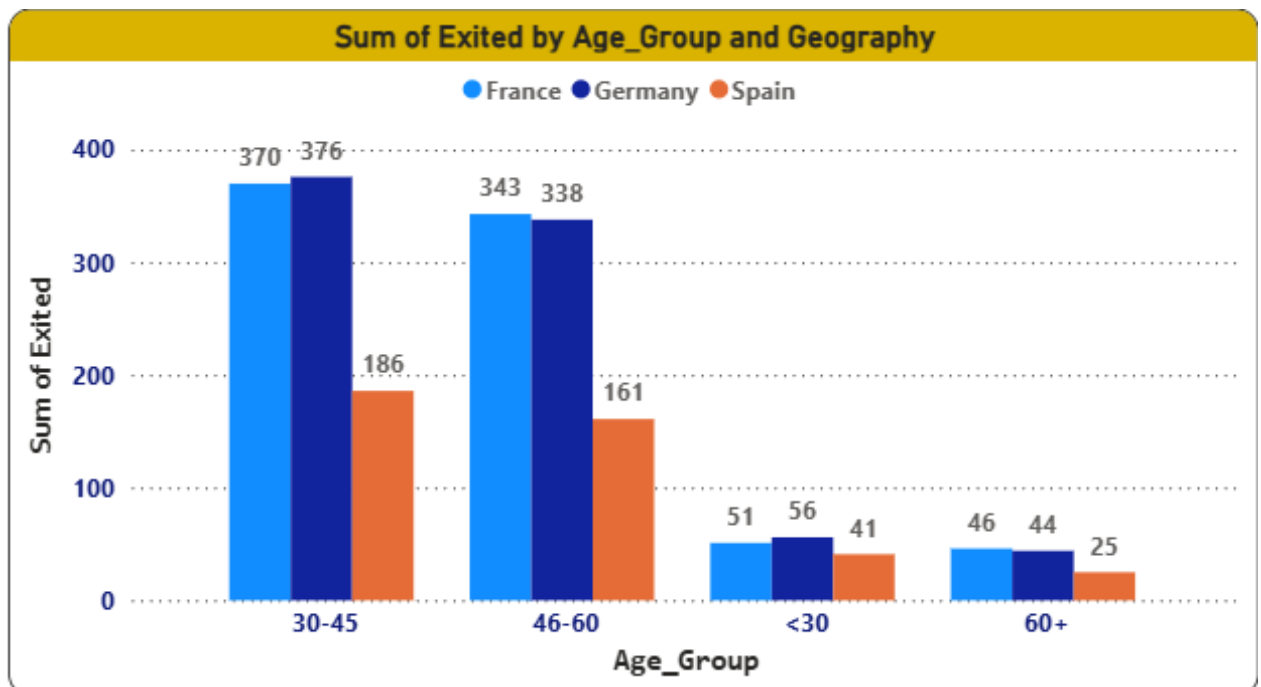
- France Total customer **5014** Exited customer **810**
- France Total Customer rate and Exited rate respectively **50.14%** and **16.16%**
- The **46-60 age group** records the highest churn rate (**42.3%**), indicating that customers approaching retirement age are more likely to leave the bank.
- The **60+ age group** also shows churn rate (**5.7%**).
- Customers aged **30-45** contribute the highest number of churned customers (**370 customers**) and have a churn rate of **45.7%**.
- Customers below **30 years** have the lowest churn rate (**6.3%**), suggesting younger customers are relatively more loyal.
- Overall, churn rates increase after age 30 and remain around **40%**, indicating that customer retention efforts should focus primarily on customers aged **30 years and above**.

○ Germany Customer Churn Analysis by Age Group :

- Germany Total customer **2509** Exited customer **814**
- Germany Total Customer rate and Exited rate respectively **25.09%** and **32.44%**
- The **46-60 age group** segment the churn rate (**41.2%**),
- The **60+ age group** shows a high churn rate (**5.4%**), although the number of customers in this segment is relatively small.
- Customers aged **30-45** contribute the highest number of churned customers (**376 customers**) and have a churn rate of **46.2%**.
- Customers below **30 years** have the lowest churn rate (**6.9%**), suggesting younger customers are relatively more loyal.

○ Spain Customer Churn Analysis by Age Group :

- Spain Total customer **2477** Exited customer **413**
- Germany Total Customer rate and Exited rate respectively **24.77%** and **16.67%**
- The **46-60 age group** records the highest churn rate (**39%**),
- The **60+ age group** shows churn rate (**6%**)
- Customers aged **30-45** contribute the highest number of churned customers (**186 customers**) and have a churn rate of **45%**.
- Customers below **30 years** have the lowest churn rate (**10%**), suggesting younger customers are relatively more loyal.



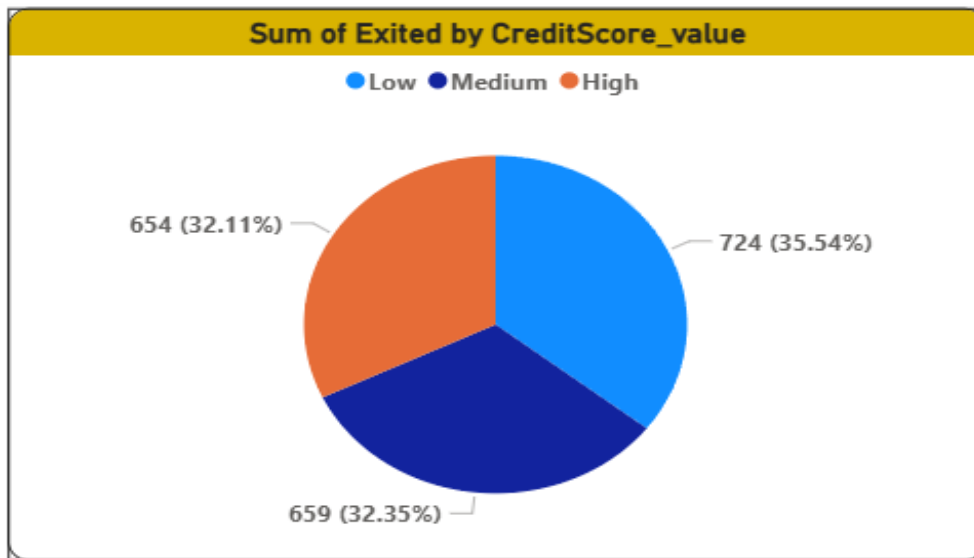
The analysis reveals that customers aged **30–60 years** are the major drivers of customer churn across all regions. Germany experiences the highest churn among customers aged **30–45**, while France experiences the highest churn among customers aged **46–60**. Therefore, the bank should prioritize customer retention strategies for middle-aged and pre-retirement customers, as these segments account for the majority of customer attrition across France, Germany, and Spain.

Credit Score Bands Segmentation :

Here see the count of exited by the credit score value.

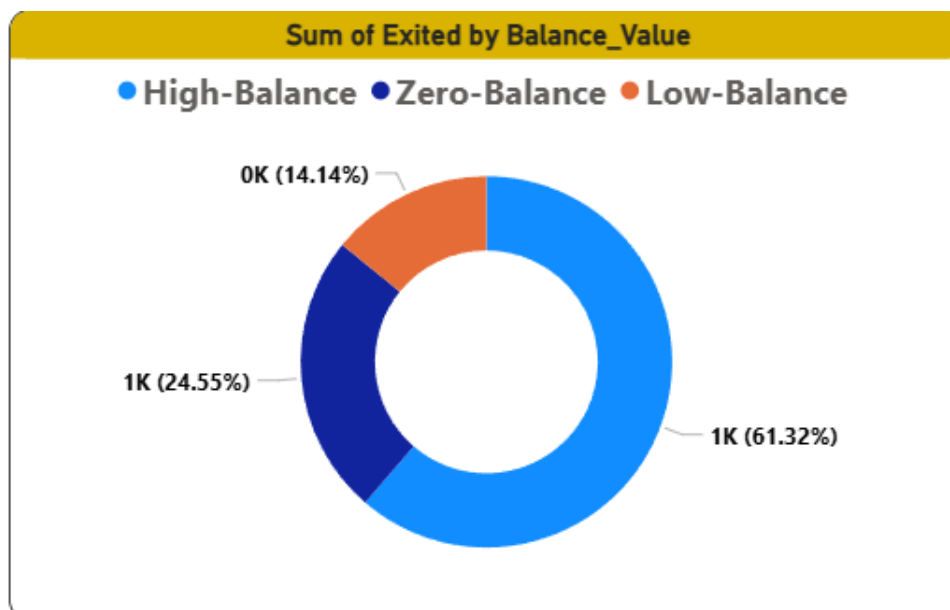
- Low credit score customers churn is **724** churn rate is **32.54%**
- Medium credit score customers churn is **659** churn rate is **32.35%**
- High credit score customers churn is **654** churn rate is **32.11%**

We can see here highest churn rate is Low Credit Score value **32.54%**



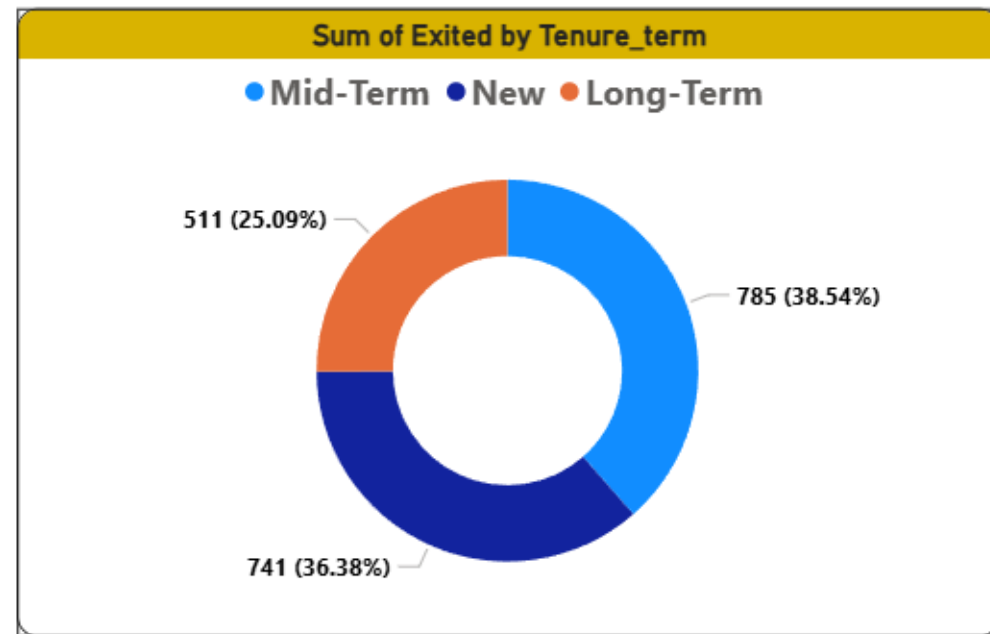
Balance Segments :

Analyze the churn pattern across to the Balance segmentation



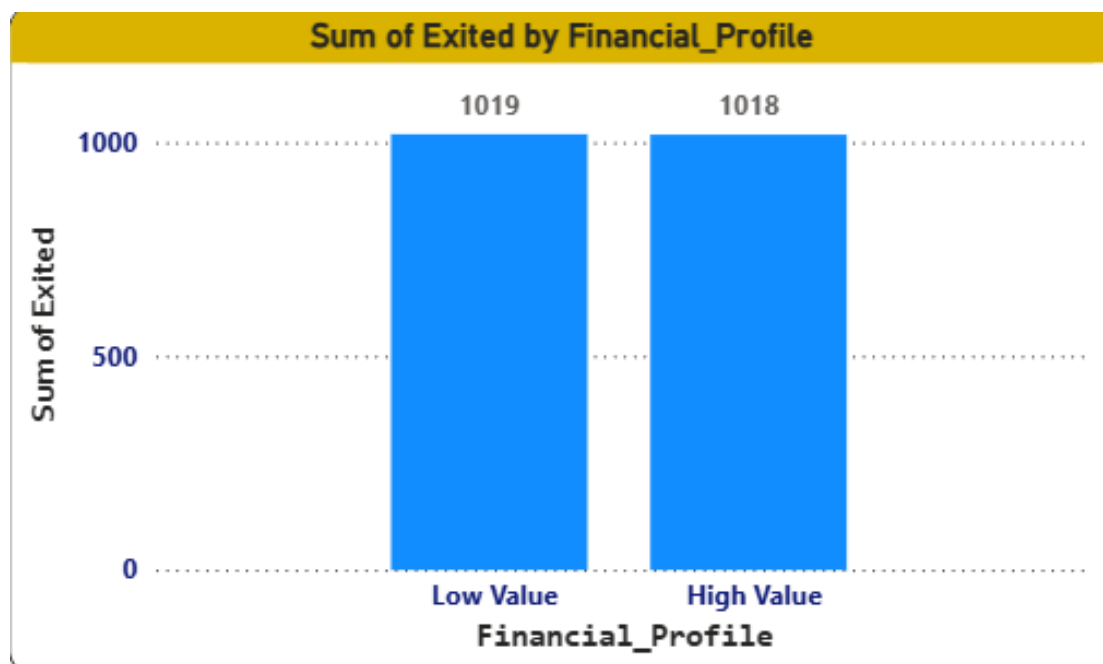
Tenure Segmentation :

Churn pattern analysing with the use of Tenure Segmentation.



Financial profile Segmentation :

The financial profile analysis examines customers' financial characteristics, including account balance, estimated salary, credit score, and the number of products owned. These factors are analyzed to understand their relationship with customer retention and churn, helping the bank identify high-risk customer segments and develop effective retention strategies.



Conclusion for financial profile segmentation

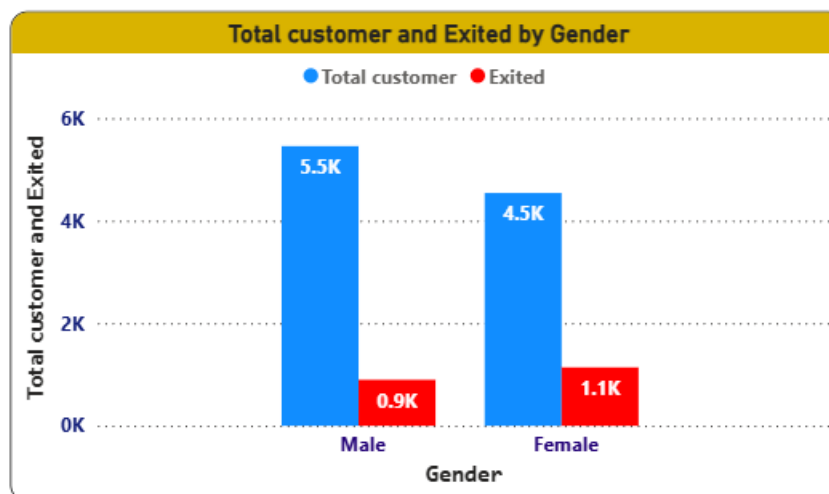
The Financial Profile analysis reveals that customer churn is almost evenly split between low-value and high-value customers, indicating that financial standing alone is not a significant predictor of customer attrition. Therefore, the bank should consider a combination of demographic, behavioural, and financial factors when designing customer retention strategies.

Gender-based churn differences :

Analyzing the churn pattern of the Gender wise. Here we are see the Gender of Male and Female in the bar chart Below.

Here,

- Total Male Customer - **5457** also their Exited Customers – **898**
 - **The Male Customers Exited rate is 16.5%**
 - **Overall Male Customers Contributing To Churn – 8.98%**
- Total Female Customer – **4543** also their Exited Customers – **1139**
 - **The Female Customers Exited rate is 25.07%**
 - **Overall Female Customers Contributing to Churn – 11.39%**



Here Females have high attrition rate in the data. So the bank should give the prioritize the Female customers for Retaining process. Bank have the high number of male customer but there male customers mostly still in the bank and theirs churn rate also low.

Key Performance Indicators (KPI) :

Over all Churn Rate – 20.37%

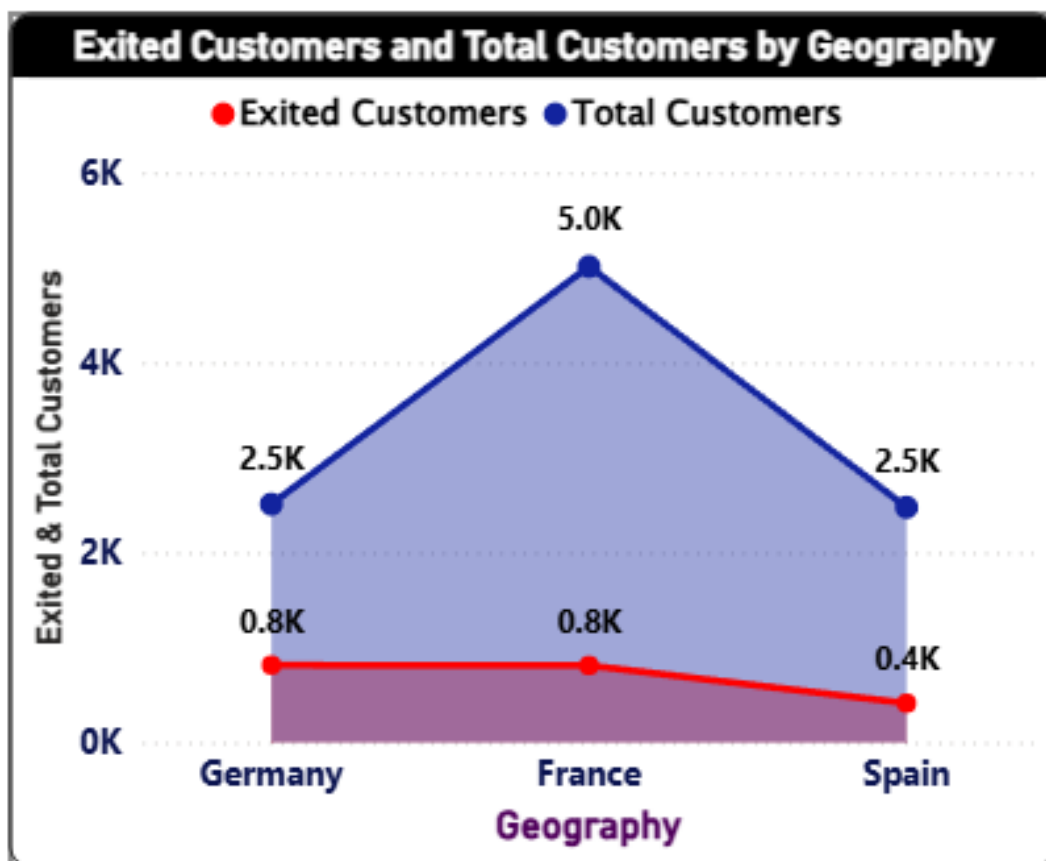
Here Already we are analysing the part of Segmentation Analysis

So Now Analyze the Churn Pattern Below,

- Geography Churn Risk pattern
- High Value Churn Ratio (the churn how to spread with low value and high value customers)
- Engagement Indicators Among the Inactive and Active Customers Churn Rate

Geography Churn Risk Pattern :

Analyz the Churn Risk Pattern with the help of Diagram. Because the diagram Explanation is gives easy to understand for Every Churn segmentation analyzing.



Now analyze the Risk pattern :

$$\text{Geographic Risk Index} = \frac{\text{Country Churn Ratio}}{\text{Overall Churn Ratio}}$$

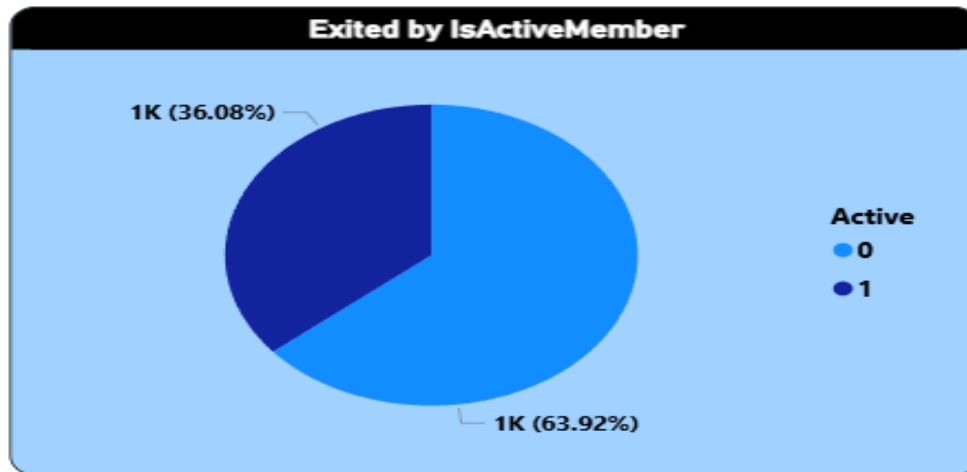
Geography	Total Customers	Exited Customers	Churn Ratio	Risk Index
France	5014	810	16.15%	0.79
Germany	2509	814	32.44%	1.59
Spain	2477	413	16.8%	0.82

Risk Index	Meaning
< 1.0	Lower-than-average churn risk
= 1.0	Average churn risk
> 1.0	Higher-than-average churn risk

Geographic Risk Analysis

- **Germany** has a Risk Index of **1.59**, meaning customers in Germany are **59% more likely to churn** than the average customer.
- **Spain** has a Risk Index of **0.82**, indicating a churn risk **18% lower** than the overall average.
- **France** has the lowest Risk Index (**0.79**), suggesting the strongest customer retention performance among the three regions.
- customer retention initiatives should be prioritized in Germany, where the likelihood of customer attrition is significantly higher than in other regions.

Engagement Drop Indicator (Inactivity vs churn) :



Here we are clearly the churn pattern spread between the Active and Inactive Customers.

Status	Total	Exited	Churn Rate
Active	5,151	735	14.3%
Inactive	4,849	1,302	26.9%

Churn Rate = $\frac{\text{Total Customers}}{\text{Exited Customers}} \times 100$

Engagement Drop Indicator Analysis

- Inactive customers exhibit a significantly higher churn rate (**26.9%**) compared to active customers (**14.3%**).
- More than half of all churned customers belong to the inactive segment.
- This indicates that customer engagement is strongly associated with customer retention.
- Customers who actively interact with the bank are less likely to leave, while inactive customers face a much greater risk of churn.

The Engagement Drop Indicator reveals that inactivity is a strong predictor of customer churn. Customers who are not actively engaged with banking services are substantially more likely to leave the bank. Therefore, improving customer engagement through personalized offers, digital banking initiatives, and loyalty programs could help reduce customer attrition.

Final Conclusion for this Churn pattern Analysis :

The analysis of the European banking customer dataset reveals that customer churn is influenced by a combination of demographic, financial, geographic, and engagement-related factors.

Key Findings

1. Overall Churn

- Approximately **20% of customers have exited the bank**, indicating that customer retention remains a significant business challenge.

2. Geographic Impact

- **Germany recorded the highest churn rate (32.44%)**, making it the highest-risk market.
- **France (16.2%)** and **Spain (16.8%)** showed considerably lower churn rates.
- Geographic location is therefore a strong predictor of customer attrition.

3. Age Influence

- Customers aged **46–60 years** exhibited the highest churn tendency.
- Younger customers (below 30) showed the strongest retention levels.
- Churn risk generally increases with age.

4. Financial Profile

- Customers with **higher account balances** leaves in the bank comparatively low-balance customers.
- Credit score showed some influence, but it was not the strongest churn driver compared to age, geography, and engagement.

5. Customer Engagement

- Inactive customers were significantly more likely to leave the bank.
- Reduced engagement emerged as one of the strongest early warning indicators of churn.

6. Tenure Analysis

- Customer tenure alone had a limited impact on churn.
- Both newer and long-term customers can leave if engagement and satisfaction decline.

Business Implications

The results suggest that customer churn is driven more by **customer engagement, age profile, geographic factors, and product usage** than by tenure or credit score alone. Banks should prioritize:

- Improving customer engagement programs.
- Targeting high-risk customers in Germany.
- Developing retention strategies for middle-aged and older customers.
- Encouraging cross-selling to increase product ownership.
- Identifying inactive customers early and implementing proactive retention campaigns.

Overall Conclusion

The European bank's churn problem is not caused by a single factor but by the interaction of demographic, behavioural, and financial characteristics. By focusing on customer engagement, geographic risk management, and stronger customer relationships through multiple product offerings, the bank can significantly reduce churn and improve long-term customer retention and profitability.